PV102 Day 5 Student Packet

Battery Storage Autonomy and Critical Loads

PV102 | CTS Day 21 | Student Print Packet
Storage has a defined job: carry named loads through named time. | Monday, August 10, 2026

Today's rule

Storage has a defined job: carry named loads through named time.

Learning targets

- Explain the difference between battery capacity, usable capacity, and autonomy.
- Estimate storage need from essential daily Wh and autonomy goal.
- Identify why battery chemistry, BMS limits, temperature, ventilation, and overcurrent protection matter.
- Defend load shedding as a design tool, not a failure.

Morning quiz

1. Battery capacity is commonly expressed as:

- A. Ah or Wh
- B. Degrees azimuth only
- C. Wire color
- D. Roof pitch only

2. A battery bank should be sized from:

- A. Defined loads and runtime
- B. The prettiest battery case
- C. Random parts available
- D. Module color

3. Usable capacity is often less than nameplate capacity because:

- A. Depth-of-discharge and protection limits matter
- B. Labels are decoration
- C. All loads are AC
- D. Sunlight is heavy

4. Autonomy means:

- A. Time the system must carry loads without enough solar input
- B. A compass direction
- C. A breaker type
- D. A module frame dimension

5. Load shedding means:

- A. Reducing or turning off lower-priority loads
- B. Cutting random wires
- C. Ignoring the client
- D. Making voltage disappear

6. Estimate: 1,000 Wh/day for 2 days = ___ Wh before losses and usable-capacity adjustment.

_____

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7. Why does battery voltage matter in system design?

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8. Name one battery safety boundary.

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9. Why is “more battery” not always the best answer?

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10. Write the Day 5 rule in your own words.

_____

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Reference: decision order

Step	Action	Proof question
1	Define the job	What must the system accomplish?
2	Collect evidence	What is measured, observed, or still unknown?
3	Choose boundary	What must be protected, isolated, sized, or verified?
4	Defend the choice	What evidence supports the decision?
5	Handoff	What should the next person know?

Critical Load Storage Plan

Teams reduce an oversized wish list into a critical-load storage plan that a real system could support.

Scenario options

- Bus conversion overnight loads: fridge, fan, lights, phone, laptop, water pump, optional induction cooktop.
- Remote sensor must survive two stormy days with no site visit.
- Small shop wants outage backup for lights, router, one outlet, and refrigerator only.

Team procedure

- Start with the Day 3 load table or provided scenario.
- Mark each load as critical, useful, or optional.
- Calculate critical daily Wh.
- Choose an autonomy target and estimate required Wh before adjustment.
- List the battery information required before final sizing.
- Handoff the critical-load list, autonomy target, and rejected optional loads.

Team worksheet

Prompt	Team response
System/job statement	
Evidence used	
Decision made	
Safety boundary	
Top unknowns	
Next action	

Exit ticket

Which load did your team remove or limit, and why did that make the system stronger?

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